MIL-S-8559A (ASG)
26 JULY 1965
Superseding
MIL-S-8559(ASG)
26 February 1953

MILITARY SPECIFICATION

STEEL BARS, CARBON, AISI (1095), AIRCRAFT QUALITY

This specification has been approved by the Department of the Air Force and by the Bureau of Naval Weapons.

1. SCOPE

1.1 Scope.- This specification covers carbon steel (1095) bars.

1.2 Classification.- Bars shall be furnished in one grade only, and in one of the following physical and surface conditions, as specified (see 6.2):

Physical conditions:

(A) As forged
(B) As rolled
(C) Spheroidized

Surface conditions:

(1) Black as forged or rolled
(2) Pickled or blast cleaned
(3) Rough turned
(4) Cold rolled or cold drawn
(5) Surface ground

1.2.1 Unless otherwise specified, bars 1 inch and less in diameter or thickness shall be furnished in condition (C) (4), and bars over 1 inch in diameter or thickness shall be furnished in condition (C) (2) (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

P-D-680	Dry Cleaning Solvent

| FSC 9510 |

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PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-C-650	Crates, Wood, Open and Covered

Military

MIL-L-7870	Lubricating Oil, General Purpose, Low Temperature
MIL-C-11796	Corrosion Preventive Compound, Petrolatum, Hot Application
MIL-C-16173	Corrosion Preventive Compound, Solvent Cutback, Cold Application

STANDARDS

Federal

Fed. Std. No. 48	Tolerances for Steel and Iron Wrought Products
Fed. Test Method Std. No. 151	Metals; Test Methods
Fed. Std. No. 183	Continuous Identification Marking of Iron and Steel Products

Military

MIL-STD-163	Steel Mill Products, Preparation for Shipment and Storage
MIL-STD-430	Macrograph Standards for Steel Bars, Billets, and Blocks

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

Society of Automotive Engineers, Inc.

AMS2301	Aircraft Quality Steel Cleanliness-- Magnetic Particle Inspection Procedure

(Copies of Aerospace Material Specifications may be obtained from the Society of Automotive Engineers, Inc., 485 Lexington Avenue, New York, N. Y. 10017.)

3. REQUIREMENTS

3.1 Material.- The steel shall be of aircraft quality. The materials shall be magnetically inspected in accordance with the procedures of AMS 2301, and shall not exceed the size and frequency rating limits indicated in the paragraph entitled "Disposition" of AMS 2301 (see 4.8).

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3.2 Manufacturing process.- The steel shall be manufactured by the open-hearth, or electric furnace process, unless a single process is specified in the contract or order.

3.2.1 Sufficient discard shall be taken from each ingot to ensure freedom from piping and undue segregation.

3.3 Chemical composition.- The chemical composition shall be as specified in table I.

TABLE I. Chemical composition

Element	Analysis (percent)	Check-analysis 1/ tolerance (percent)
Carbon	0.90 - 1.05	±0.03
Manganese	0.30 - 0.50	±0.03
Phosphorus	0.040 (max) 2/	+0.008
Sulfur	0.050 (max)	+0.008

1/ The average of all the separate determinations made shall be within the limits specified in the "Analysis" column. Individual determinations may vary to the extent shown in the "Tolerance" column, except that the several determinations of a single element in any one heat shall not vary both above and below the specified range.

2/ Maximum of 0.05 permitted when acid steel is specified.

3.4 Hardness limits for condition (C).- Material furnished in condition (C) shall have a maximum hardness of Brinell 207 (Rockwell B-95). The hardness of material in condition (C) (4) shall not exceed Brinell 229.

3.5 Grain size.- The grain size shall be predominantly No. 5 or finer, with grains as large as No. 3 permissible.

3.6 Macrostructure.- Visual examination of deep acid etched sections shall be equal or superior to plates A2, B2, and C1 of MIL-STD-430. Plates D1 through D8 are unacceptable.

3.7 Tolerance.-

3.7.1 Diameter or thickness.- The permissible variation in dimensions of the bars shall be as shown on Federal Standard No. 48, except that when intended for re-forging purposes, the requirements of Federal Standard No. 48 are waived.

3.8 Lengths.-

3.8.1 Exact lengths.- Bars may be ordered to exact lengths or in lengths expressed as a multiple of a definite unit, with tolerances as specified in the contract or order.

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3.8.2 Mill lengths.- When exact or multiple lengths are not ordered, bars will be accepted in mill lengths of 6 to 20 feet, but not more than 10 percent of any order shall be furnished in lengths shorter than 10 feet.

3.9 Identification of product.- When specified, or for direct shipments to the Government, each bar 1/2 inch or more in diameter or 3/8 inch or greater in width, shall be legibly marked throughout its length with a row of continuously recurring symbols in accordance with Federal Standard No. 183. The symbols shall be applied with a marking fluid which is not affected by the solvents used in MIL-C-11796, class 1; MIL-C-16173, grade 1; water, or solvent conforming to P-D-680. The gap between recurring legends shall not exceed 2 feet. The marking used shall not rub off or be smeared by contact incident to normal handling during shipment or storage. The following information shall be included:

MIL-S-8559
Condition
Manufacturer's name or trademark

3.9.1 Cold finished bars.- All bars smaller than 1/2 inch in diameter or 3/8 inch in width of flat shall be bundled and tagged at each end with the information listed in 3.9, using an oilproof tag, with an extra tag included in the bundle.

3.9.2 Hot rolled bars.- Hot rolled bars, 1 inch and over in diameter or width and with the surface conditions indicated in the specification, shall be marked with the specification number on or near each end of the bar. Hot rolled bars under 1 inch shall be bundled. Bundles should be tagged at each end and an extra tag included in the bundle.

3.10 Workmanship.- Bars shall be sound, commercially straight, of uniform quality and condition, free from pipes, laps, cracks, twists, seams, damaged ends, or other injurious defects, and shall have a smooth finish of the best quality. Bars shall be commercially machinable and shall be capable of taking a good finish.

3.10.1 Condition (4) bars shall be free from scale or other surface imperfections.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- The examination and testing of the steel shall be classified as quality conformance inspection.

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4.3 Examinations.-

4.3.1 Examination of product.- Each bar shall be carefully examined to determine conformance with this specification with respect to tolerances, identification, workmanship, surface condition, and dimensions.

4.3.2 Preservation, packaging, and packing.- Preparation for delivery shall be examined for conformance to section 5.

4.4 Chemical analysis.-

4.4.1 Sampling.- One or more samples for chemical analysis shall be selected in accordance with Federal Test Method Standard No. 151, to represent each heat of steel. The sample shall consist of not less than 2 ounces.

4.4.1.1 Location.- Samples for chemical analysis shall be taken parallel to the axis of the bar selected, at an area midway between the center and surface, except that material less than 1-1/4 inches thick shall be sampled through the entire cross section.

4.4.1.2 Waiver.- Samples for chemical analysis may be waived provided that all of the material under inspection can be identified as being made from a heat previously analyzed and found to be in conformance with the chemical composition specified herein.

4.4.1.3 Unidentified material.- Where the material is not identified in accordance with 3.9 by the producer, or the identity of any portion of the shipment is obscure in any respect, sampling shall be in accordance with table II.

TABLE II. Sampling

Lot size	Sample size	Acceptance No.
1 to 7	All	0
8 to 40	7	0
41 to 110	15	1
111 to 180	25	2
181 to 301	35	3
Over 301	50	4

4.4.2 Test methods.- Samples shall be prepared in accordance with Fed. Test Method Std. No. 151, and shall be tested by wet chemical, spectrographic, or other analytical methods. In the event of dispute, analysis shall be by wet chemical methods.

4.5 Hardness limits for condition (C).-

4.5.1 Sampling.- Sampling shall be in accordance with table II.

4.5.2 Preparation of specimen.- Hardness test specimens shall conform to Fed. Test Method Std. No. 151.

4.5.3 Method.- Hardness tests shall be conducted in accordance with Fed. Test Method Std. No. 151.

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4.6 Grain size determination.-

4.6.1 Sampling.- One or more representative samples from each heat in the shipment shall be selected for the determination of grain size.

4.6.2 Method.- Specimens shall be sectioned and polished to appropriate fineness by metallographic methods and suitably etched to reveal the grain structure. The austenitic grain size shall be determined by Procedure C or D of Fed. Test Method Std. No. 151.

4.7 Macrostructure (deep acid etch).-

4.7.1 Sampling.- At least two samples shall be selected from each heat of steel for examination of macrostructure (deep acid etch).

4.7.2 Preparation of specimens.- Specimens shall be cut from the ends of the bars selected and shall represent the entire cross section of the bar. The specimen shall measure 1/2 inch or more in the direction of the axis of the bar. One of the faces of the specimen representing the cross section of the bar shall be finished flat and smooth by a fine machine cut or by grinding. The finished face of the specimen shall be etched in an aqueous solution containing 50 percent hydrochloric acid by volume and maintained at a temperature of approximately 71° C (160° F).

4.7.3 Method.- Specimens shall be examined by a competent metallographist for the presence of defects.

4.8 Magnetic inspection quality.- The specimens shall be selected, inspected, and rated in accordance with the procedures of AMS 2301. Tests may be made on semi-finished mill products.

4.9 Rejection and retest.- Where failure of a specimen is definitely ascribed to faulty material or fails to meet the applicable test requirements, the entire lot shall be rejected. Where failure of any lot of material to meet the requirements of this specification is due to inadequate heat treatment, the material may be reheat treated and resubmitted for test. Only one such reheat treatment will be allowed.

5. PREPARATION FOR DELIVERY

5.1 Level A.- Materials shall be prepared for shipment in accordance with MIL-STD-163. Materials with surface conditions (2) through (5) shall be coated with preservatives as prescribed for condition (4).

5.2 Level B.-

5.2.1 Preservation.- Bars furnished in surface conditions (2) to (5), inclusive, shall be coated with oil conforming to MIL-L-7870.

5.2.2 Packaging.- Materials preserved as specified in 5.2.1 and those in surface condition (1), shall be properly separated by size and condition when packed for shipment and shall be prepared for shipment in units convenient to handle at destination.

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5.2.3 Packing.- Unless otherwise specified, materials in surface conditions (2) to (5), inclusive, shall be packed in wooden boxes or crates. Material furnished in surface condition (1) shall be packed in secured lifts or as specified for conditions (2) to (5). All exterior shipping containers shall be so constructed as to ensure acceptance by common or other carrier for safe transportation at the lowest rate to the point of delivery. The weight of exterior shipping containers or bundles shall not exceed 2,000 pounds. Containers shall be able to withstand rehandling and reshipment without the necessity of repacking.

5.3 Marking of shipment.- Marking and labeling shall be in accordance with MIL-STD-163.

6. NOTES

6.1 Intended use.- The steel is intended for use in the manufacture of parts subjected to high shear loads or extensive wearing. Such applications include steel balls, pins, keys, springs, etc.

6.2 Ordering data.- Procurement documents should specify:

(a) Title, number, and date of this specification.
(b) Thickness, diameter, or shape and physical and surface conditions (see 1.2).
(c) Level of preparation for delivery (see section 5).

Custodians:
Air Force - (11)
Navy - WP

Preparing activity:
Navy - WP

Reviewer activities:
Air Force - (11)
Navy - WP

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget
Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION MIL-S-8559A(ASG) Steel Bars, Carbon, AISI (1095), Aircraft Quality

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ Direct Government Contract ☐ Subcontract

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)

SUBMITTED BY (Printed or typed name and activity) DATE

DD Form 1426

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OFFICIAL BUSINESS

Aeronautical Standards Group
8719 Colesville Road
Silver Spring, Maryland 20910

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